Controlling two-qubit involutions: CNOT costs and symbolic invariants

Gregory Morse[1] and Péter Rakyta[2,3]

[1]Department of Programming Languages and Compilers, Eötvös Loránd University, Hungary
[2]Department of Physics of Complex Systems, Eötvös Loránd University, Hungary
[3]Computational Sciences Department, HUN-REN Wigner Research Centre for Physics, Hungary
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We classify the ancilla-free CNOT cost of controlling a traceless Hermitian involution on two qubits, allowing arbitrary one-qubit unitaries. The minimum is four at operator-Schmidt rank four, three at rank two, and one or two at rank one according to local support. A control-space rigidity argument excludes three arbitrary-angle Ising interactions in the rank-four case. Controlled fermionic SWAP attains this bound with a literal four-CNOT, six-rotation circuit, whereas three unrestricted two-qubit gates suffice. We extend the normal-form analysis to every spectral stratum of two-qubit involutions. Trace-± 2 reflections form a one-Schmidt-angle family joining controlled-controlled-Z and Fredkin: we give exact seven-CNOT and five-unrestricted-gate upper bounds and partial lower bounds in its interior. Finally, we reformulate the known two-SQiSW region as a necessary-and-sufficient real polynomial test in matrix entries, suitable for symbolic families. The accompanying symbolic library, phase-resolved circuits, and row-level certificates provide reproducible constructions with explicit optimality scopes. Detailed compiler experiments and numerical gap audits are supplied separately.

1 Introduction and related work

Controlling an operator preserves more structure than its ordinary two-qubit local-equivalence class. In particular, local factors in a target block become conditional operations. We exploit this observation to classify a continuous class of three-qubit gates: $C(V) = \mathrm{diag}(I_4, V)$, where $V = V^\dagger$, $V^2 = I_4$, and $\mathrm{tr}\, V = 0$. A two-angle normal form gives matching constructions on every operator-Schmidt-rank stratum. A separate control-space argument excludes three Ising interactions on the rank-four stratum. The resulting classification is the main contribution.

Hermitian unitaries are observables with eigenvalues ± 1; controlling one is the central operation in a Hadamard test of its expectation value. With a $|+\rangle$ control, the probability of the final $+1$ outcome is $(1 + \langle V \rangle)/2$. Controlled Pauli products are elementary examples; locally conjugated observables and two-qubit reflections give less trivial instances. Controlled oracles and reflections also occur in linear-combination and qubitization constructions [1]. Our question is the exact interaction cost of such a small controlled block. It does not require or imply a complete algorithmic speedup. Spectral multiplicity separates the balanced two-angle family from the one-angle rank-one-reflection family containing ordinary Fredkin and controlled-controlled-Z.

Gate-count lower bounds based on operator-Schmidt strength originate in work including Nielsen et al. [2]. Yu, Duan and Ying proved that Toffoli requires five unrestricted two-qubit gates [3]; Yu and Ying classified controlled-controlled gates and proved the five-gate unrestricted Fredkin bound [4]. Their controlled-product arguments are directly relevant here. Shende and Markov studied block-diagonal three-qubit circuits with fewer than six CNOTs and proved Toffoli's six-CNOT minimum [5]. Our Ising proof uses the same general principle of restricting intermediate control frames, with an explicit Schmidt-space rigidity lemma. Balanced involutions exclude the nonbalanced relative blocks of ordinary Fredkin and Toffoli; our smaller counts concern a different class.

Two-qubit invariant and Cartan methods are established [6, 7]. The universal two-B construction is due to Zhang et al. [8]. Huang et al. give analytical SQiSW synthesis, its two-pulse Weyl region, and its universal three-pulse bound [9]. Our second result is an real polynomial reformulation of that known region, with an exact algebraic proof and implementation. Peterson, Crooks, and Smith characterize fixed-depth two-qubit reachability as monodromy polytopes [10]. The trace-invariant small-CNOT tests of Shende, Bullock, and Markov [7] are the closest methodological analogue. The benefit here is a test on symbolic matrix entries without solving an eigenvalue problem; for a numerical 4×4 matrix, ordinary canonicalization is already inexpensive. Our symbolic charts implement these established pulse counts with scalar tracking and explicit degeneracy branches; the counts and the existence of analytical local frames are prior results.

The accompanying software joins OSR factorization, native analytical charts, and exact local rewrites. Quartz [11] and QUESO [12] already synthesize verified, gate-set-dependent optimizations. UniversalQCompiler provides analytical synthesis in Mathematica [13]. Single-qubit exact Clifford+T synthesis and normal forms have an extensive literature [14, 15, 16, 17]; the 192 phase-resolved one-qubit Clifford circuits in our atlas are a small reference and regression suite. Neither their number nor an easily reproducible shortest-path search is presented as a major synthesis advance. Earlier numerical OSR partition resynthesis [18] is a separate project and supplies no numerical evidence for the identities proved here.

2 Resources, phase, and operator-Schmidt rank

All gate-count theorems use arbitrary one-qubit unitaries, all-to-all connectivity, fixed input/output wires, arbitrary input states, and no ancillas. An Ising interaction is locally equivalent to e^{itZZ}; its angle may vary between uses. CNOT is one member of this class. An unrestricted two-qubit gate is a different, strictly larger resource. For supplied hardware alphabets, the archive counts the declared primitive; a theorem in the relaxed model transfers as a lower bound, and requires a separate native construction to prove a matching optimum.

Literal synthesis requires $U = C$. Projective synthesis requires

$$U = e^{i\gamma} C, \qquad |e^{i\gamma}| = 1, \tag{1}$$

with the scalar recorded at zero physical cost. A physical GP gate is never inserted. At equal declared cost we prefer literal equality. Controlling a projective word requires $P(\gamma)$ on the added control, since $C(U) = P_c(\gamma) C(C)$. The release proves a concrete controlled-Sycamore substitution both with and without this correction. This illustrates the contract; SDKs that track global phase can already implement this rule.

For a cut $A|B$, realign $U_{ab,a'b'}$ with row (a, a') and column (b, b'). The rank is $r_{A|B}(U) =$

$\mathrm{OSR}_{A|B}(U)$. It obeys

$$r_{A|B}(UV) \le r_{A|B}(U) r_{A|B}(V), \qquad r_{A|B}(L_A \otimes L_B) = 1. \tag{2}$$

For a controlled operator, its *control OSR* means this rank on $c|ab$. If its two target branches are linearly independent it is two, and its control-side Schmidt space is the full diagonal space $\mathcal{D} = \mathrm{span}\{|0\rangle\langle 0|, |1\rangle\langle 1|\}$. For a target cut, the V block is a submatrix of the realignment of $C(V)$. Thus $\mathrm{OSR}(V) = 4$ forces ranks $(2, 4, 4)$ on the three singleton cuts. Every crossing Ising interaction has rank at most two, so Eq. (2) constrains the number and placement of interactions.

3 A rank-four controlled-block lower bound

Theorem 1 *If* $\mathrm{OSR}(V) = 4$*, then* $C(V)$ *requires at least four Ising-class interactions, even with arbitrary interaction angles, arbitrary one-qubit unitaries, and an arbitrary scalar. Controlled fermionic SWAP attains this bound with four CNOTs.*

3.1 Control-space rigidity

Lemma 1 *If invertible one-qubit matrices* P, Q *map the full diagonal matrix space*

$$\mathcal{D} = \mathrm{span}\{|0\rangle\langle 0|, |1\rangle\langle 1|\}$$

onto itself by $D \mapsto PDQ$*, then* P, Q *are monomial with the corresponding common permutation. Thus locally equivalent controlled operators with control OSR two have corresponding conditional target blocks, up to phases, exchange of the two branches, and common target-local input/output frames.*

Proof. The only rank-one lines in $\mathcal{D}$ are its two coordinate projectors: $\det \mathrm{diag}(a, b) = ab$. An invertible left/right map preserves rank and must permute these two lines. Its column and row factors therefore have the stated form. Control Schmidt spaces of rank-two controlled operators are full copies of $\mathcal{D}$, proving the second claim. In particular, a locally equivalent realization of $\mathrm{diag}(\mathbb{I}, V)$ has a local conditional branch. Its relative target block has the same OSR as V, also after branch inversion. *QED*

3.2 Exclusion of three interactions

Proof of the lower bound in Theorem 1. Call the targets a, b. The singleton cuts have ranks $2, 4, 4$ for c, a, b: the rank-four assertion follows from the V block, since a target cut has maximum rank four. Each crossing Ising interaction has rank at most two. Consequently a and b each have degree at least two and c at least one in an interaction graph. Two interactions cannot satisfy these demands. With three, the only edge multisets are a triangle ca, cb, ab, or two ab edges and one control edge. The triangle has six time orders. The doubled-target-edge case has three placements of the control edge for each of its two possible targets, also six. These are the twelve surviving edge words.

First consider one control crossing and two internal target interactions. Absorb local frames into adjacent gates and put the single crossing in controlled form. Its target branches are W_+W_- and W_+QW_-, where Q is a one-wire unitary and $W_\pm$ contain the target operations after/before the crossing. By Lemma 1, choose a local branch as the first one (inverting Q if the branches are exchanged). If both internal interactions lie on one side, the corresponding product must be local, so the relative branch is local. If

one lies on each side, the relative branch is $W_-^\dagger Q W_-$ with W_- locally Ising. To make the conjugation bound explicit, write the one-wire operator as $q_0 I + q_x X + q_y Y + q_z Z$. Conjugating by e^{itZZ} leaves I and Z unchanged, sends $X \otimes I$ to a combination of $X \otimes I$ and $Y \otimes Z$, and sends $Y \otimes I$ to a combination of $Y \otimes I$ and $X \otimes Z$. Consequently the result is $M_0 \otimes I + M_1 \otimes Z$ and has OSR at most two. Outer local frames preserve that rank. In neither case can the relative block have rank four.

Now consider the triangle. Put each control interaction in controlled form and combine all operations on the control between them into a one-qubit unitary N. Local frames at the endpoints are harmless by Lemma 1. A 2×2 unitary is either monomial or has all four entries nonzero. If N is monomial, the circuit remains controlled after a possible control permutation. Each conditional target branch contains the sole internal ab interaction, with only local factors around it. That interaction must be entangling; otherwise the singleton rank demands already fail. Neither branch is local, a contradiction.

If N has all entries nonzero, factor the target operation K between the crossings to one side. The four control blocks then span $\{\mathbb{I}, B, A, AB\}$, where B is a nontrivial one-wire unitary on one target and A is the other controlled target action conjugated through K. Their span dimension equals the control OSR, namely two. If the internal interaction is outside the crossings, K is local, and A, B act nontrivially on different wires. The four tensor products are independent, contradicting rank two. If the internal interaction lies between them, independence of $\mathbb{I}, B$ forces $A \in \mathrm{span}\{\mathbb{I}, B\}$. The transformed controlled gates then act only on c, b. Apart from local factors the whole circuit is $(\mathbb{I}_c \otimes K_{ab}) W_{cb}$. Its $a|cb$ rank is at most $\mathrm{OSR}(K) = 2$, again a contradiction. These cases exclude every three-interaction word. The four-CNOT identity in Sec. 4.3 attains the bound for U_F. *QED*

The supplied checker verifies the exact matrix identity, cut minors, three-edge enumeration, Ising conjugation identities, and native-class identifications. The argument above is an analytical proof with checked algebraic components, not a proof-assistant formalization or a conclusion from failed numerical optimization.

4 Complete classification for balanced involutions

Theorem 2 *Let $V = V^\dagger$, $V^2 = I_4$, and $\mathrm{tr}\, V = 0$. The minimum CNOT and unrestricted two-qubit-gate counts of $C(V)$ are:*

Relative block	*CNOT*	*Unrestricted*
$\mathrm{OSR}(V) = 4$	*4*	*3*
$\mathrm{OSR}(V) = 2$	*3*	*3*
$\mathrm{OSR}(V) = 1$, *one-body Pauli up to local conjugacy*	*1*	*1*
$\mathrm{OSR}(V) = 1$, *two-body Pauli product up to local conjugacy*	*2*	*2*

The CNOT lower bounds use Theorem 1 in the rank-four row and the unrestricted two-gate exclusion below in the rank-two row. There is no rank-three stratum. Multiplying V by an arbitrary scalar phase does not change these interaction counts when arbitrary $U(2)$ locals are available: the scalar is supplied by a physical one-qubit phase on c.

4.1 Two-angle normal form, including degeneracies

Expand a traceless Hermitian V in the real Pauli basis:

$$V = \vec{a} \cdot \vec{\sigma} \otimes I + I \otimes \vec{b} \cdot \vec{\sigma} + \sum_{ij} T_{ij} \sigma_i \otimes \sigma_j. \tag{3}$$

Multiplying Pauli matrices and setting $V^2 = I$ gives exactly

$$|a|^2 + |b|^2 + \|T\|_F^2 = 1, \quad Tb = 0, \quad T^T a = 0, \quad \mathrm{cof}(T) = ab^T. \tag{4}$$

The scalar, one-body, and two-body coefficients of V^2 independently verify these equations. The supplied checker expands every coefficient.

If a, b are nonzero, rotate them to the X and Z axes respectively. Then the X row and Z column of T vanish. Its remaining real 2×2 block has a signed singular-value decomposition under the remaining rotations in the two perpendicular planes. Signs are retained, so only proper rotations, implemented by local unitaries, are needed. This gives

$$V_0 = AX_a + BZ_b + CY_aY_b + DZ_aX_b, \quad A^2 + B^2 + C^2 + D^2 = 1, \quad AB + CD = 0. \tag{5}$$

For completeness, if only a is nonzero, $\mathrm{cof}(T) = 0$ forces $\mathrm{rank}\, T \leq 1$ and $T^T a = 0$ makes its nonzero left singular axis perpendicular to a. The two remaining terms are put into Eq. (5) by local rotations. The case with only b is symmetric. If both vanish, T has rank one and unit norm, giving a Pauli product. If $T = 0$, exactly one of a, b is nonzero and has unit norm. Thus the same normal form covers every degenerate case.

The matrix $\left(\begin{smallmatrix} A & -C \\ D & B \end{smallmatrix}\right)$ has rank one and Frobenius norm one, so it is an outer product of two real unit vectors. Write these as $(\sin\alpha, \cos\alpha)^T$ and $(\sin\beta, \cos\beta)$. With $K = \mathrm{CX}_{a,b}$ and $Q(t) = R_y(t) Z R_y(-t)$ this proves

$$V_0 = K\,[Q(\alpha)_a Q(\beta)_b]\,K, \tag{6}$$

with coefficients $(A, B, C, D) = (\sin\alpha\sin\beta, \cos\alpha\cos\beta, -\sin\alpha\cos\beta, \cos\alpha\sin\beta)$. In the Pauli coefficient matrix the four coefficients occupy different rows and columns. Hence OSR is their number of nonzero entries, which is one, two, or four. The relation $AB+CD = 0$ excludes three.

4.2 Matching constructions and two-gate exclusions

Controlling Eq. (6) uses two outer K gates and one controlled Q on each target. Since every Q is a one-qubit Pauli axis, its controlled gate is locally CZ and costs one CNOT. This proves the four-CNOT upper bound throughout the class. Local conjugating frames for V add only one-qubit gates to $C(V)$.

On rank-two boundaries there is a stronger form $V_0 = K' Q_j K'^{\dagger}$ with K' containing one CNOT. The table specifies every angle; $S = \mathrm{diag}(1, i)$ and matrix products act right to left.

Edge	t	K'	Q_j
$\alpha = 0$	β	$H_b \mathrm{CX}_{a,b}$	$\cos t X_b + \sin t Z_b$
$\alpha = \pi/2$	β	$H_a S_b \mathrm{CX}_{a,b}$	$\sin t Z_a + \cos t Y_a$
$\beta = 0$	α	$S_a^\dagger \mathrm{CX}_{b,a}$	$\cos t Z_b + \sin t Y_b$
$\beta = \pi/2$	α	$H_a \mathrm{CX}_{a,b}$	$\sin t Z_a + \cos t X_a$

Other edges follow by angle periodicity and local Pauli conjugations. The matrix identity is $C(V_0) = K' C(Q_j) K'^{\dagger}$: in *time order* apply $K'^{\dagger}$, then $C(Q_j)$, then K'. This costs three CNOTs. The checker proves all four identities with t free, including the corners.

To exclude two unrestricted interactions for nonlocal V, their edges must cover all three wires. If both edges meet c, the controlled-product argument of Yu and Ying [4, Proposition 2] says that the conditional branches are target-local; their relative product cannot be nonlocal. If the edges are ca, ab, there is a single crossing of the control cut.

Its OSR must be two. A two-qubit unitary of OSR two is locally controlled: in its Cartan expansion, rank two leaves only one nonzero interaction coordinate. Put this gate into controlled form. With the ab interaction entirely on one side of the crossing, the local-branch argument of Lemma 1 makes the relative target branch local. The reversed order and exchanged targets give the same conclusion. Two gates therefore cannot implement any nonlocal relative V.

For any balanced V, its $+1$ and -1 eigenspaces both have dimension two. An unrestricted two-qubit change of basis J satisfies $V = JZ_aJ^\dagger$. The circuit $J^\dagger, CZ_{ca}, J$, in time order, uses three unrestricted interactions. Together with the preceding exclusion and Theorem 1, this proves both nonlocal rows of Theorem 2. In the rank-one cases, local Hermitian unitary factors reduce to identities or Pauli axes. One-body control costs one CX; a two-body Pauli product costs two. One gate cannot cover its three nontrivial singleton cuts, proving the latter minimum.

The theorem depends on more than Weyl class. The relative blocks I, Z_a, and Z_aZ_b all have the local Weyl class, whereas their controlled costs are zero, one, and two. A classification of arbitrary controlled-V by the Weyl class of V alone is therefore impossible. The present theorem classifies balanced involutions, not all controlled unitaries.

4.3 Controlled fermionic SWAP

Let $F|00\rangle = |00\rangle$, $F|01\rangle = |10\rangle$, $F|10\rangle = |01\rangle$, and $F|11\rangle = -|11\rangle$. Then $F = (Z_a + Z_b + X_aX_b + Y_aY_b)/2$, so its four independent Pauli Schmidt coefficients give $\mathrm{OSR}(F) = 4$. Equivalently, $F = (S^\dagger \otimes S^\dagger)\mathrm{iSWAP}$. It is Hermitian, squares to identity, and has trace zero. Ordinary SWAP has trace two and belongs to a different class.

Figure 1 is a literal identity with four CNOTs and six R_y rotations. In time order its rotations and interactions are

$$R_y(-\pi/2)_a,\ \mathrm{CX}_{a,b},\ R_y(-\pi/4)_a,\ R_y(-3\pi/4)_b,$$
$$\mathrm{CX}_{c,a},\ \mathrm{CX}_{c,b},\ R_y(\pi/4)_a,\ R_y(3\pi/4)_b,\ \mathrm{CX}_{a,b},\ R_y(\pi/2)_a.$$

Exact multiplication gives $C(F)$ with no leftover scalar. Six is a local-count upper bound, not a proved local minimum. An independently constructed two-qubit J, rotating the odd-parity subspace by a real $\pi/4$ rotation, also gives $F = J^\dagger Z_a J$; its exact three-interaction controlled circuit is checked separately.

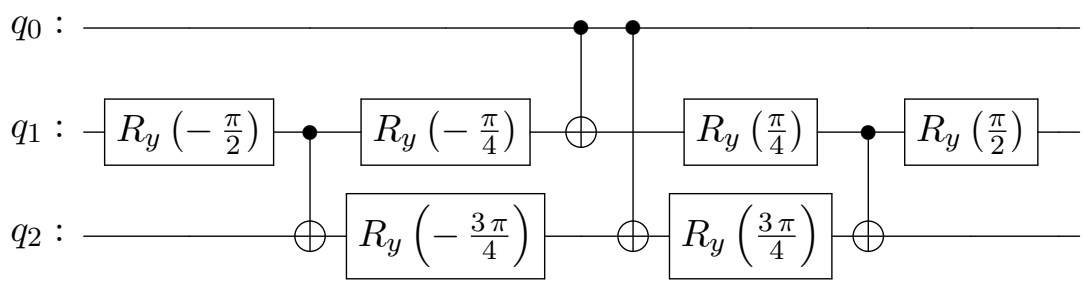


Figure 1: Literal controlled fermionic SWAP. Wire zero is c, and wires one and two are a, b. The interaction count is optimal in the Ising model.

5 The remaining spectral strata

For a two-qubit Hermitian involution the trace is $0, \pm 2$, or ± 4. The balanced stratum was classified above. Trace ± 4 gives $V = \pm I$, so $C(V)$ is identity or Z_c and needs no interactions. The remaining strata are reflections about a single two-qubit state, up to sign.

Proposition 1 *Every trace-± 2 two-qubit involution is locally conjugate to $\pm R_\theta$, where*

$$R_\theta = I - 2|\psi_\theta\rangle\langle\psi_\theta|, \qquad |\psi_\theta\rangle = \cos\theta|00\rangle + \sin\theta|11\rangle, \quad 0 \le \theta \le \pi/4. \tag{7}$$

Every $C(\pm R_\theta)$ has an ancilla-free literal construction with at most seven CNOTs, or five unrestricted two-qubit gates. For $0 < \theta < \pi/4$,

$$4 \le E_{\mathrm{CX}}(C(R_\theta)) \le 7, \qquad 3 \le E_{\mathrm{arb}}(C(R_\theta)) \le 5. \tag{8}$$

At $\theta = 0$ the respective exact minima are six and five. At $\theta = \pi/4$ the unrestricted minimum is five, and the CNOT bounds are $5 \le E_{\mathrm{CX}} \le 7$. The interior bounds do not assert a cost classification.

Proof. The spectral theorem writes a trace-two involution as $I - 2|\psi\rangle\langle\psi|$. Schmidt decomposition puts $|\psi\rangle$ into the stated form using local conjugations, with phases absorbed in the local frames. Trace minus two changes only the sign. Controlling that sign adds Z_c.

Let $J_\theta = \mathrm{CX}_{a,b} R_y(2\theta)_a$. Since $J_\theta|00\rangle = |\psi_\theta\rangle$,

$$C(R_\theta) = J_\theta (X_a X_b)\, CCZ_{c,a,b}\, (X_a X_b) J_\theta^\dagger. \tag{9}$$

A six-CNOT CCZ word plus the two outer target CNOTs gives an initial bound of eight. The improved seven-CNOT word is given, in full time order, in the supplement. Its only varying local angles are $\pi - 2\theta$ and $2\pi - 2\theta$ in two U_3 gates; all constants are rational multiples of π. With $U_3(\delta, \phi, \lambda) = P(\phi) R_y(\delta) P(\lambda)$, direct multiplication yields $\mathrm{diag}(I_4, R_\theta)$, with

$$R_\theta = \begin{pmatrix} -\cos 2\theta & 0 & 0 & -\sin 2\theta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\sin 2\theta & 0 & 0 & \cos 2\theta \end{pmatrix}.$$

The circuit uses seven CNOTs and eighteen physical U_3 gates, without a GP gate or residual scalar. For the improved unrestricted count put

$$B_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ -i & 1 & 0 & 0 \\ i & 1 & 0 & 0 \end{pmatrix}, \quad \widetilde{J}_\theta = \mathrm{CX}_{b,a} R_y(2\theta)_b, \quad B_\theta = B_0 \widetilde{J}_\theta^\dagger.$$

The matrix basis is $|ba\rangle$, with a the least significant target wire. All factors are explicitly unitary. Since X_a commutes with $\widetilde{J}_\theta$,

$$B_\theta R_\theta X_a B_\theta^\dagger = Z_a S_b = \mathrm{diag}(1, -1, i, -i).$$

Consequently the following five gates, in time order, implement $C(R_\theta)$:

$$\mathrm{CX}_{c,a}, \quad B_{\theta,ab}, \quad CS_{c,b}, \quad CZ_{c,a}, \quad B_{\theta,ab}^\dagger. \tag{10}$$

The zero-control branch is $B_\theta^\dagger B_\theta = I$, and the one-control branch is $B_\theta^\dagger Z_a S_b B_\theta X_a = R_\theta$. The release proves this literal identity with θ free.

Direct realignment of R_θ gives

$$\det \mathcal{R}(R_\theta) = -\sin^2(2\theta)[2 - \sin^2(2\theta)], \tag{11}$$

which is nonzero on $(0, \pi/4]$. Theorem 1 therefore excludes three CNOTs, and the preceding general two-gate exclusion gives the unrestricted lower bound three. At zero, R_0 is a negative-control CZ; its controlled gate is locally CCZ. Shende–Markov's six-CNOT and Yu–Duan–Ying's five-unrestricted-gate bounds apply [5, 3]. At $\pi/4$, a local Pauli frame maps $|\psi_\theta\rangle$ to the singlet. Reflection about the singlet is SWAP, so this endpoint is locally Fredkin. Yu–Ying gives its unrestricted minimum five [4]; seven CNOTs is the best known construction count in our literature audit, and supplies the stated upper bound [19]. Seven is not asserted to be a proved minimum. *QED*

Thus all spectral strata have explicit normal forms and constructions. Only the balanced stratum has a complete interaction-cost classification; the interpolation between the Toffoli and Fredkin endpoints remains a concrete optimization problem. In particular, the five-gate word proves a uniform upper bound, not a five-gate lower bound in the interior.

6 An invariant characterization of the two-SQiSW region

Let $\Gamma = U(YY)U^T(YY)$ and $\Delta = \det U$. Define the scalar-independent Makhlin invariants [6]

$$g_1 = \frac{(\operatorname{tr}\Gamma)^2}{16\Delta}, \qquad g_2 = \frac{(\operatorname{tr}\Gamma)^2 - \operatorname{tr}(\Gamma^2)}{4\Delta}. \tag{12}$$

Use $\mathcal{A}(x, y, z) = \exp[i(xXX + yYY + zZZ)]$, with chamber $\pi/4 \geq x \geq y \geq |z|$, and SQiSW $= \mathcal{A}(\pi/8, \pi/8, 0)$.

Proposition 2 *A two-qubit unitary has a projective realization with at most two SQiSW pulses and arbitrary local unitaries if and only if*

$$W = g_2 + 1 \geq 0, \qquad W^2 \geq 8(|g_1| + \Re g_1). \tag{13}$$

This is an invariant rewriting of Huang et al.'s region $x \geq y + |z|$ [9], including its boundary.

Proof. In the chamber, $g_1 = (u + iv)^2$ up to the immaterial sign of v, where $u = \cos 2x \cos 2y \cos 2z \geq 0$, and $g_2 = \cos 4x + \cos 4y + \cos 4z$. Thus $|g_1| + \Re g_1 = 2u^2$, and Eq. (13) is equivalent to $W \geq 4u$. Put $X = \cos 2x$, $Y = \cos 2y$, $Z = \cos 2|z|$. Direct factorization gives

$$\begin{aligned} W - 4u &= 2(X^2 + Y^2 + Z^2 - 2XYZ - 1) \\ &= 2[X - \cos 2(y + |z|)][X - \cos 2(y - |z|)]. \end{aligned} \tag{14}$$

When $|z| > 0$, the second factor is negative: $x \geq y > y - |z|$ and cosine decreases on the relevant interval. The first factor is nonpositive exactly when $x \geq y + |z|$. When $z = 0$, the product becomes $(X - Y)^2 \geq 0$ and the chamber already satisfies $x \geq y$. Huang et al.'s necessary-and-sufficient chamber characterization now proves both directions. Exact polynomial checks establish Eq. (14) and the identity for $(u+iv)^2$, independently of random sampling. *QED*

For a directly polynomial version put $a = \Re g_1$, $b = \Im g_1$ and $W = g_2 + 1$. Equation (13) is equivalent to

$$W \geq 0, \qquad W^2 - 8a \geq 0, \qquad W^4 - 16aW^2 - 64b^2 \geq 0. \tag{15}$$

Indeed $W^2 - 8a \geq 8\sqrt{a^2 + b^2}$ is equivalent to its nonnegative left-hand side and the squared inequality. For unitary U, $1/\det U = \overline{\det U}$, so these are real polynomial inequalities in the real and imaginary matrix entries, with constant denominators. They can be substituted into symbolic families while retaining parameter conditions; a numerical canonicalization advantage is not claimed.

The test decides zero/one/two-pulse-region membership, not the exact minimum within that region. A separate zero/one-pulse class test is needed to assert that two pulses are necessary. No new universal three-pulse bound is claimed.

7 Software and data

The implementation combines exact OSR factorization, native analytical charts, and local rewrites, accepting identities or compositional exact certificates. Its 327×32 reference grid contains 217 named one-qubit entries (including 192 phase-resolved Cliffords), 90 other named entries, and 20 fixed register fixtures. These are catalog entries rather than independent nonlocal families. The supplement reports coverage by physical qubit count, obstruction type, and matched proof scope, with a row-level optimality ledger. The finite Clifford+T shortest-word certificates are a regression/reference resource alongside established synthesis theory [14, 15, 16, 17].

An exact monodromy audit now certifies the minimum pulse counts of all 64 fixed-target/native pairs in the audited SYC and square-root-SWAP columns. Other records certify named parameter samples. Exact rational primal and Farkas certificates decide membership and exclusion; exact complete invariants certify the coordinate conversion [10, 6]. A count theorem can establish existence of a shorter word while its explicit symbolic local frames remain unfinished. Those two facts are displayed separately in the atlas. The exact controlled-phase SYC chart now uses two native pulses, and DCX has an explicit one-TK2 word. The atlas also flags remaining shorter *numerical* witnesses in the SYC and $\sqrt{\mathrm{SWAP}}$ columns, separately from its displayed exact circuits. Parameter-family samples are labelled as samples. The detailed compiler comparisons, chart/OSR ablations, parameter-reuse timings, and negative peephole experiment are in the supplement. They measure the supplied implementation and do not establish compiler dominance or a general runtime advantage.

The supporting software, exact words, proof-component checks, audit circuits, and ledger are supplied at `https://github.com/rakytap/sequential-quantum-gate-decomposer`, with research sources pinned to tag `symbolic-involutions-research-20261108`, commit `26912300eef3ea11fe94eea05e5b23f5b8180d98`. The exact matrix checks support the analytical proofs. The row-specific rational certificates are logically required for the enumerated native-count claims; numerical compiler runs are supporting experiments and regression evidence, not proof premises. Separate atlas and scientific-data archives retain editable LaTeX, complete records, and content-hash manifests.

8 Scope and open questions

The balanced classification allows arbitrary one-qubit unitaries, all-to-all connectivity, and no ancillas. A restricted local alphabet needs its own native construction. The Ising lower bound permits arbitrary interaction angles but does not apply to unrestricted two-qubit gates. The trace-± 2 interior has partial bounds; determining its exact costs is open. We conjecture that the proved upper bounds, five unrestricted gates and seven CNOTs, are both attained as minima for every $0 < \theta \leq \pi/4$. Both upper constructions now have

literal free-angle identities; the missing interior lower bounds remain open. Earlier numerical witnesses are retained for provenance and are not used as lower-bound evidence. The Fredkin endpoint likewise has a best known seven-CNOT word, with no seven-CNOT minimum claimed here. Continuous local-cost minima and all-grid interaction optimality are also open, and twenty shared-expression scores remain unevaluated. Numerical witnesses supply optimization leads, not exact identities or impossibility proofs. We infer no calibrated pulse latency or fidelity gain from local-box counts. The analytical proofs have exact checked components rather than proof-assistant formalizations.

Author contributions. Gregory Morse: conceptualization, computer-science methodology and analysis, software, mathematical analysis, validation, experiments, data curation, visualization, and original draft. Péter Rakyta's planned contributions are physics review and interpretation of the results, scientific discussions, and manuscript drafting, reviewing, and editing.

Use of AI. OpenAI GPT-6.1 Sol (xhigh reasoning mode) assisted research, proof development and checking, calculations, software implementation, experiments, literature and citation research, and manuscript drafting and revision. Anthropic Claude Opus 5.5 (high reasoning mode) provided iterative simulated referee reports. The authors are responsible for the mathematics, software, data, and final text.

References

[1] Guang Hao Low and Isaac L. Chuang. Hamiltonian Simulation by Qubitization. *Quantum*, 3:163, 2019. DOI: 10.22331/q-2019-07-12-163.

[2] Michael A. Nielsen, Christopher M. Dawson, Jennifer L. Dodd, Alexei Gilchrist, Duncan Mortimer, Tobias J. Osborne, Michael J. Bremner, Aram W. Harrow, and Andrew Hines. Quantum dynamics as a physical resource. *Physical Review A*, 67:052301, 2003. DOI: 10.1103/PhysRevA.67.052301. URL `https://arxiv.org/abs/quant-ph/0208077`.

[3] Nengkun Yu, Runyao Duan, and Mingsheng Ying. Five two-qubit gates are necessary for implementing the Toffoli gate. *Physical Review A*, 88(1):010304, 2013. DOI: 10.1103/physreva.88.010304.

[4] Nengkun Yu and Mingsheng Ying. Optimal simulation of Deutsch gates and the Fredkin gate. *Physical Review A*, 91(3):032302, 2015. DOI: 10.1103/physreva.91.032302.

[5] Vivek V. Shende and Igor L. Markov. On the CNOT-cost of TOFFOLI gates. *Quantum Information and Computation*, 9(5-6):461–486, 2009. DOI: 10.26421/QIC9.5-6-8. URL `https://arxiv.org/abs/0803.2316`.

[6] Yuriy Makhlin. Nonlocal Properties of Two-Qubit Gates and Mixed States, and the Optimization of Quantum Computations. *Quantum Information Processing*, 1(4): 243–252, 2002. DOI: 10.1023/a:1022144002391.

[7] Vivek V. Shende, Stephen S. Bullock, and Igor L. Markov. Recognizing small-circuit structure in two-qubit operators. *Physical Review A*, 70(1):012310, 2004. DOI: 10.1103/physreva.70.012310.

[8] Jun Zhang, Jiri Vala, Shankar Sastry, and K. Birgitta Whaley. Minimum Construction of Two-Qubit Quantum Operations. *Physical Review Letters*, 93(2):020502, 2004. DOI: 10.1103/physrevlett.93.020502.

[9] Cupjin Huang, Tenghui Wang, Feng Wu, Dawei Ding, Qi Ye, Linghang Kong, Fang Zhang, Xiaotong Ni, Zhijun Song, Yaoyun Shi, Hui-Hai Zhao, Chunqing Deng, and

Jianxin Chen. Quantum Instruction Set Design for Performance. *Physical Review Letters*, 130(7):070601, 2023. DOI: 10.1103/physrevlett.130.070601.

[10] Eric C. Peterson, Gavin E. Crooks, and Robert S. Smith. Fixed-Depth Two-Qubit Circuits and the Monodromy Polytope. *Quantum*, 4:247, 2020. DOI: 10.22331/q-2020-03-26-247. URL https://arxiv.org/abs/1904.10541v4. Corrected arXiv version v4, 2021: increasing/decreasing conventions and Figure 6.

[11] Mingkuan Xu, Zikun Li, Oded Padon, Sina Lin, Jessica Pointing, Auguste Hirth, Henry Ma, Jens Palsberg, Alex Aiken, Umut A. Acar, and Zhihao Jia. Quartz: Superoptimization of Quantum Circuits. In *Proceedings of the 43rd ACM SIGPLAN International Conference on Programming Language Design and Implementation*, 2022. DOI: 10.1145/3519939.3523433. URL https://arxiv.org/abs/2204.09033.

[12] Amanda Xu, Abtin Molavi, Lauren Pick, Swamit Tannu, and Aws Albarghouthi. Synthesizing Quantum-Circuit Optimizers. *Proceedings of the ACM on Programming Languages*, 7(PLDI):140, 2023. DOI: 10.1145/3591254. URL https://arxiv.org/abs/2211.09691.

[13] Raban Iten, Oliver Reardon-Smith, Emanuel Malvetti, Luca Mondada, Gabrielle Pauvert, Ethan Redmond, Ravjot Singh Kohli, and Roger Colbeck. Introduction to UniversalQCompiler, 2021. URL https://arxiv.org/abs/1904.01072. Preprint first posted 2019; revised 2021.

[14] Ken Matsumoto and Kazuyuki Amano. Representation of Quantum Circuits with Clifford and $\pi/8$ Gates, 2008. URL https://arxiv.org/abs/0806.3834.

[15] Vadym Kliuchnikov, Dmitri Maslov, and Michele Mosca. Fast and efficient exact synthesis of single-qubit unitaries generated by Clifford and T gates. *Quantum Information and Computation*, 13(7–8):607–630, 2013. DOI: 10.26421/qic13.7-8-4.

[16] Brett Giles and Peter Selinger. Exact synthesis of multiqubit Clifford+T circuits. *Physical Review A*, 87(3):032332, 2013. DOI: 10.1103/physreva.87.032332.

[17] Brett Giles and Peter Selinger. Remarks on Matsumoto and Amano's normal form for single-qubit Clifford+T operators, 2019. URL https://arxiv.org/abs/1312.6584.

[18] Gregory Morse, Muhammad Al Farizi, Jakab Nádori, László Hajas, Tamás Kozsik, and Peter Rakyta. Entanglement-Aware Quantum Circuit Optimization through Globally Optimal Partitioning, 2026. URL https://arxiv.org/abs/2611.03845.

[19] Pedro M. Q. Cruz and Bruno Murta. Shallow unitary decompositions of quantum Fredkin and Toffoli gates for connectivity-aware equivalent circuit averaging. *APL Quantum*, 1:016105, 2024. DOI: 10.1063/5.0187026. URL https://arxiv.org/abs/2305.18128.